ARTIN (ALIREZA) KAVOOSI

M.Sc. Candidate, System Modeling and Data Analytics University of Tehran

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PROFILE

M.Sc. candidate in System Modeling and Data Analytics at University of Tehran, working on decision-making under uncertainty, reinforcement learning, and optimization for perishable and safety-critical supply chains. Research spans dynamic pricing, inventory control, and combinatorial optimization, with emphasis on methods that are theoretically grounded and useful in practice. Developer of open-source RL and optimization visualizations.

RESEARCH INTERESTS

Methods: Markov Decision Processes, Deep Reinforcement Learning, Linear & Integer Programming, Robust Optimization, Bandit Algorithms, Cross-Entropy Method, Neural Ordinary Differential Equations, Structural Causal Graph Dynamics

Domains: Healthcare Operations, Perishable Inventory Control, Supply Chain Coordination & Competition, Supply Chain Resilience, Dynamic Pricing, Combinatorial Optimization

EDUCATION

University of Tehran — Tehran, Iran

M.Sc. in System Modeling and Data Analytics, School of Industrial Engineering (Expected) — *Sep 2023 – Dec 2026*

- GPA: 18.62/20 (4.0/4.0) | Supervisor: Prof. Reza Tavakkoli-Moghaddam
- Thesis: "Towards Autonomous Supply Chains for Perishable Products: Learning-Based Decision-Making for Coordination, Competition, and Resilience"
- Focus: Deep RL for perishable inventory, decentralized RL with causal dynamics, duopoly competition with lead times

Kharazmi University — Tehran, Iran

B.S. in Industrial Engineering — *Sep 2019 – Jun 2023*

- GPA: 16.44/20

PUBLICATIONS

Published Journal Papers

1. **Kavoosi, A.**, Tavakkoli-Moghaddam, R., Sajedi, H., Tajik, N., & Tafakkori, K. (2025). Dynamic Pricing and Inventory Control of Perishable Products by a Deep Reinforcement Learning Algorithm. *Expert Systems with Applications*, 291, 128570. [DOI] [Project Page]

Preprints

1. **Kavoosi, A.** (2026). Stochastic Queens Elimination. SSRN Working Paper. [SSRN] [Code] [Note] [Interactive Simulator]

Working Papers Under Review / Submitted

1. **Kavoosi, A.**, Tavakkoli-Moghaddam, R., Ziari, M., & Azaron, A. Coordinating Pricing and Replenishment in a Blood Supply Chain with Deep Reinforcement Learning. *Under review.*
2. **Kavoosi, A.**, Tavakkoli-Moghaddam, R. Disruption-Resilient Blood Supply Chain by a Structural Causal Graph Dynamics and Decentralized Reinforcement Learning Approach. *Under review.*
3. **Kavoosi, A.**, Tavakkoli-Moghaddam, R., Dolgui, A. Cross-Entropy Neural Networks for Mixed Strategies in a Finite-Horizon Supplier–Retailer Inventory Duopoly with Lead Times and Rationing.

Submitted.

4. **Kavoosi, A.**, Tavakkoli-Moghaddam, R. Online Learning of Neural Ordinary Differential Equations for Dynamic Inventory Control under Non-Stationary Demand. *Submitted.*

FEATURED PROJECTS & OPEN SOURCE

Stochastic Queens Elimination (SQEP) 2026

Finite-horizon Markov decision processes for stochastic n-queens variant

- Developed adaptive coverage greedy with worst-case approximation guarantee and exact mixed integer programming for deterministic case.
- Built interactive policy simulator and visualizations. Paper | Code | Interactive | Counterexample Note

Deep Reinforcement Learning for Perishable Pricing & Inventory Control 2025

Joint dynamic pricing and inventory control

- Deep reinforcement learning approach published in Expert Systems with Applications. Focus on perishability and non-stationary demand.
- Published Paper

OR Visualizations 2026

From-scratch visualization series for metaheuristics

- Three-part series: Genetic Algorithms, Ant Colony Optimization (Elitist Ant System on 17-city TSP), Simulated Annealing on deceptive landscape.
- Designed to make search dynamics visible and verifiable against exact solutions. Read Series | GA Code | Max-Cut on a Sphere | GA Evolution

INTERACTIVE RESEARCH & BLOG NOTES

- **An interactive counterexample to greedy-degree** Sep 2026 Two queens tie for highest degree on 4-queen board, yet one is exactly optimal while other never recovers. [Interactive Note]
- **Simulated Annealing** Aug 2026 A cooling search on deceptive energy landscape. [Note]
- **Ant Colony Optimization** Jul 2026 From-scratch Elitist Ant System on 17-city TSP small enough to solve exactly. [Note]
- **Genetic Algorithm Evolution** Real-valued GA searches Rastrigin function. [Interactive]
- **Max-Cut on a Sphere** A graph, a sphere, and one random plane. [Note]
- Full list: akavosi.github.io/blog/

TALKS & PRESENTATIONS

- **MDP Seminar, University of Tehran** Feb 2025 Invited seminar on MDPs, RL literature, Bellman optimality equation. [Slides] [Post]
- **2nd Conference on Future Directions for Industrial Engineers** May 2024 Invited talk for UTIESA (University of Tehran Industrial Engineering Scientific Association). Presented in Farsi. [Slides - Farsi]

ACADEMIC AND RESEARCH EXPERIENCE

Research Assistant Sep 2024 – Present

University of Tehran, Dept. of Industrial Engineering *Tehran, Iran*

- Research on inventory control and supply chain problems under supervision of Prof. Tavakkoli-Moghaddam. Projects include blood supply chain coordination with deep RL, resilience with causal dynamics, duopoly with rationing, Neural ODEs for non-stationary demand.

Teaching Assistant Mar 2024 – Sep 2025
University of Tehran *Tehran, Iran*

- Machine Learning (Spring 2025, Prof. Jamili), Macroeconomics (Spring 2025, Prof. Ghaderi), Soft Computing (Fall 2024, Prof. Tavakkoli-Moghaddam), Linear Algebra (Spring 2024, Prof. Rahbari)

Teaching Assistant Mar 2023 – Aug 2023
Kharazmi University *Tehran, Iran*

- Engineering Mathematics II (Spring 2023, Prof. Khamse)

WORK EXPERIENCE

Business Partner Jun 2020 – Present
Family Fashion Boutiques (Aylar & Tirajeh) *Tehran Province, Iran*

- Co-manage three family-owned boutiques. Oversee inventory management, merchandising, sales. Practical retail operations experience informing research on perishable/fashion inventory control.

Research Intern Apr 2022 – Oct 2022
SAPCO – Supplying Automotive Parts Company (Iran Khodro Group) *Tehran, Iran*

- Data-driven transportation cost estimation, SAP-based inventory optimization, BPMN workflow analysis. SAPCO manages sourcing for Iran Khodro, one of Iran's largest automotive manufacturers.

AWARDS AND HONORS

- **Top 0.4% among 10,000+** University Master's Degree Entrance Exam (Konkour), Industrial Engineering, Iran, 2023
- **Top 1% among 150,000+** Nationwide University Entrance Exam (Konkour), Mathematics & Physics, Iran, 2019

ACADEMIC SERVICE

- Reviewer: *Transportation Research Part E: Logistics and Transportation Review*
- Open-source contributor: RL implementations, optimization algorithms, scientific visualizations github.com/akavosi

SKILLS

- **Programming:** Python (PyTorch, Gymnasium, NumPy, Matplotlib), Julia, GAMS, LaTeX, IBM SPSS
- **Research Software:** Develop open-source implementations of optimization algorithms, RL methods, scientific visualization, computational experiments
- **Languages:** Persian (Native), Turkish (Native), English (Professional proficiency academic writing & presentations)

REFERENCES

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